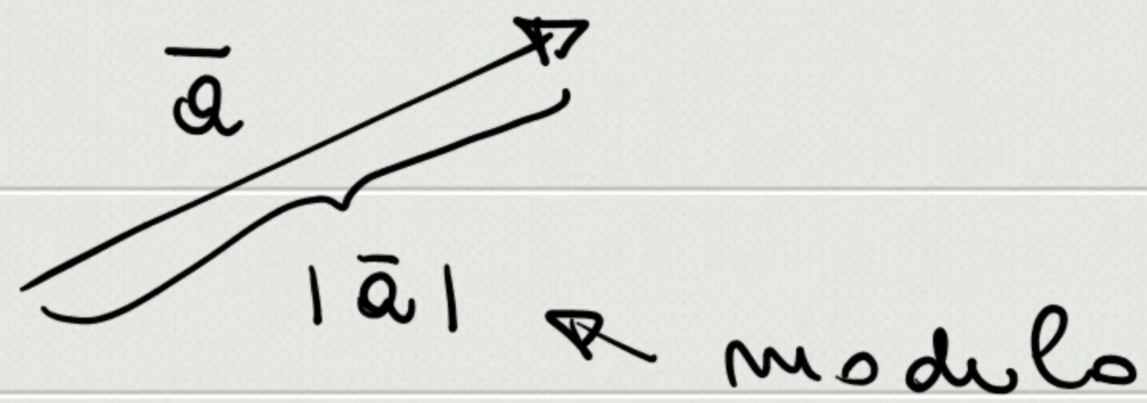
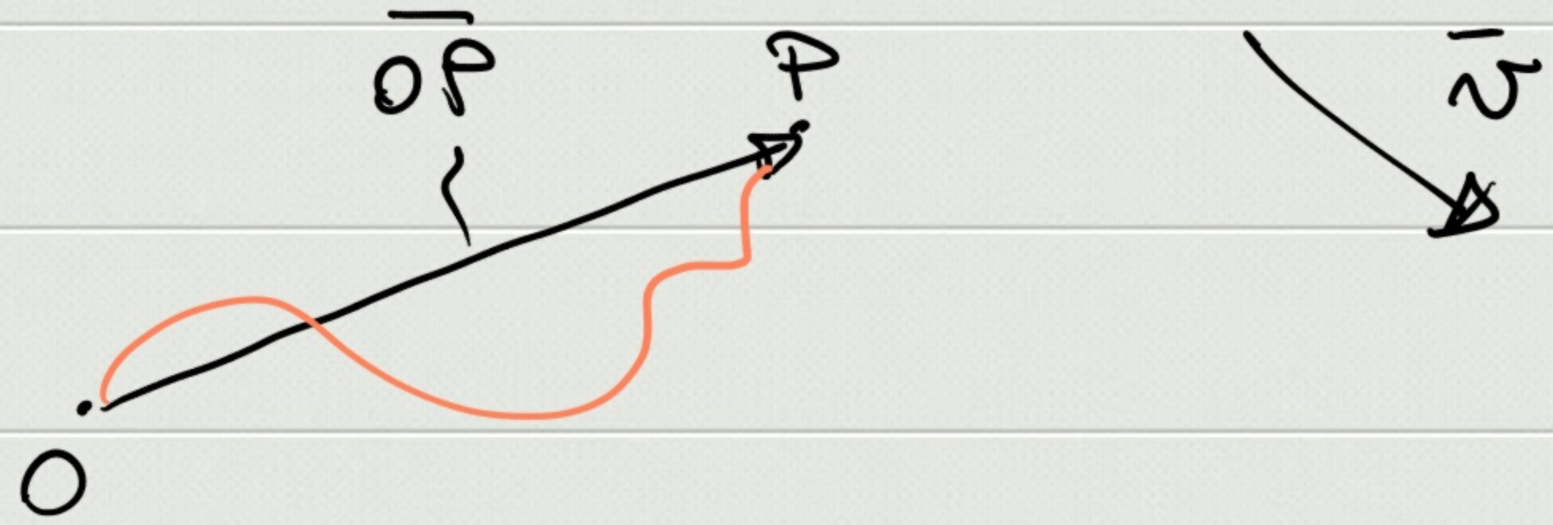
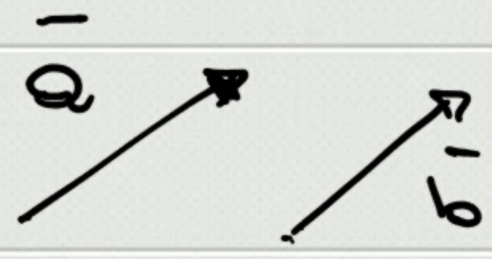
Grandezze $\begin{cases} \text{scalari} : \underline{\text{numero}} \\ \text{vettoriali} : \begin{cases} \text{numero} \\ \text{direzione} \\ \text{verso} \end{cases} \end{cases}$



$\vec{a}$, $\underline{a}$ ...



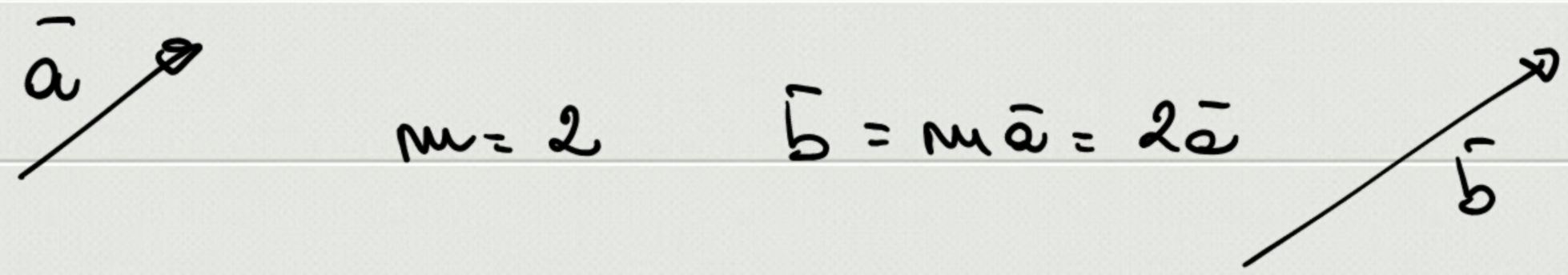


$\bar{a} = \bar{b}$ ("equipollenti")

$\bar{a}$, m (scalare) $\Rightarrow \boxed{\bar{b} = m\bar{a}}$

$\bar{b}$: $|\bar{b}| = |m\bar{a}| = |m||\bar{a}|$

$m \gtrless 0$ $\qquad$ $m < 0 \Rightarrow \bar{b}$ verso opposto $\bar{a}$



$m = 2$ $\qquad$ $\bar{b} = m\bar{a} = 2\bar{a}$

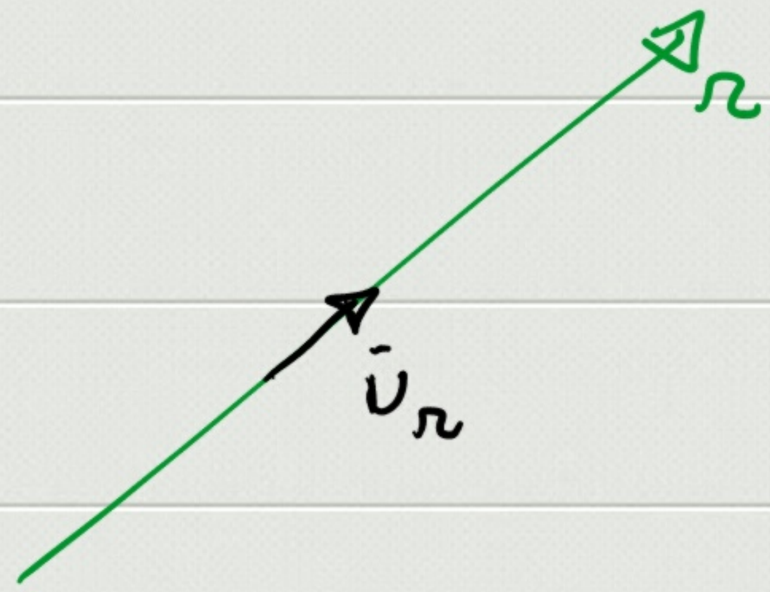
$m = -\frac{1}{2}$ $\qquad$ $\bar{b}$

$\bar{b} = -\bar{a}$: $m = -1$ (vettore opposto)

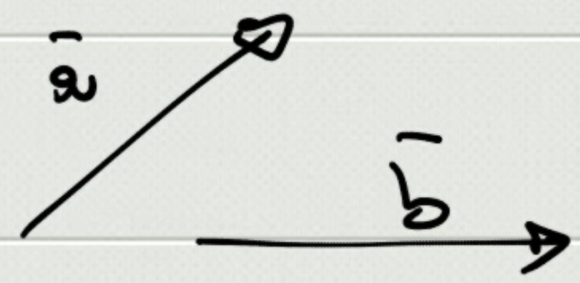
$m = \frac{1}{|\bar{a}|} \Rightarrow \bar{b} = m\bar{a} = \frac{1}{|\bar{a}|}\bar{a}$

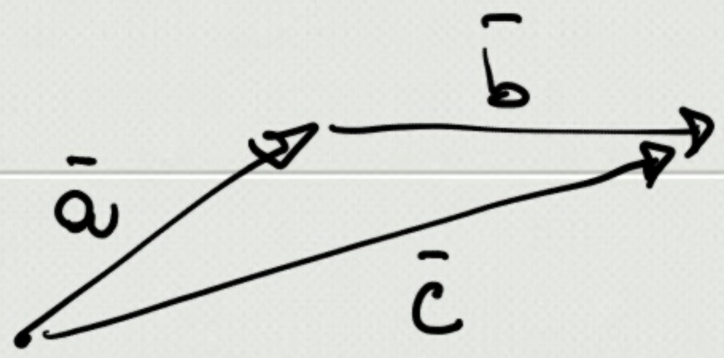
$$|\bar{b}| = \frac{1}{|\bar{a}|}|\bar{a}| = 1$$

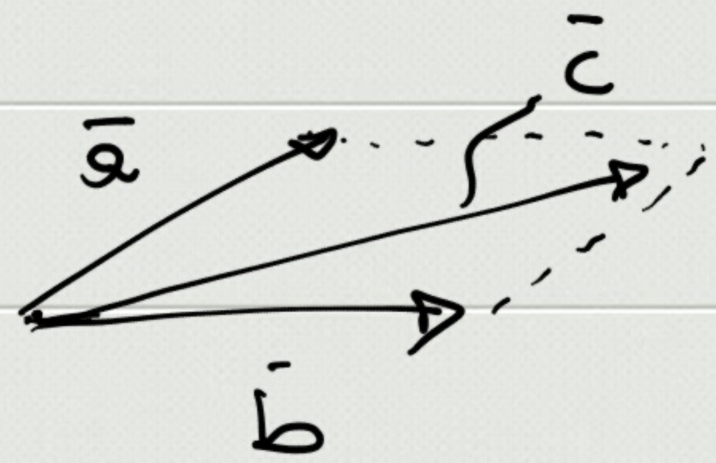
versore : vettore di modulo unitario : $\bar{u}$



- $\bar{c} = \bar{a} + \bar{b}$

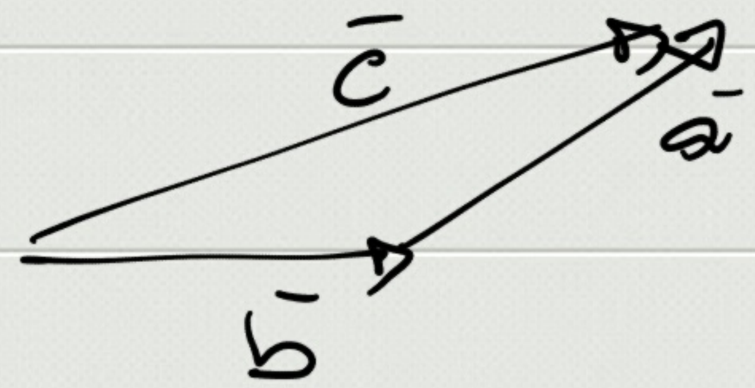




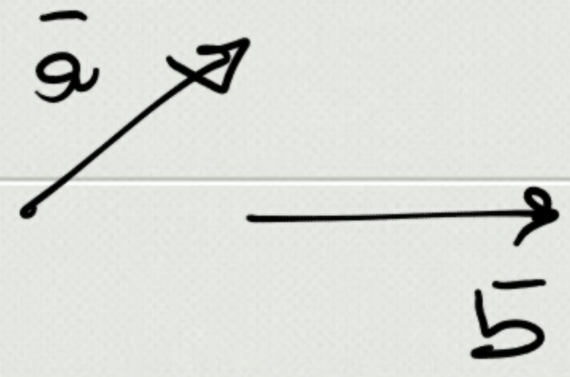


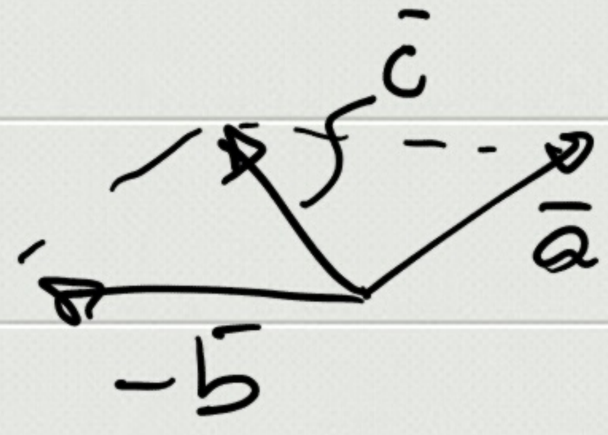
$\cancel{c = a + b}$

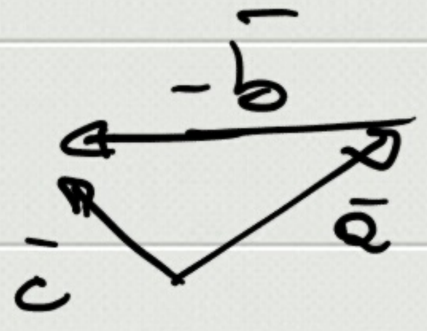
$\bar{a} + \bar{b} = \bar{b} + \bar{a}$

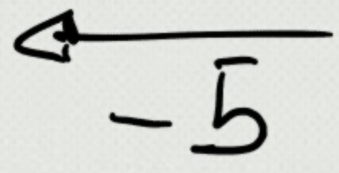


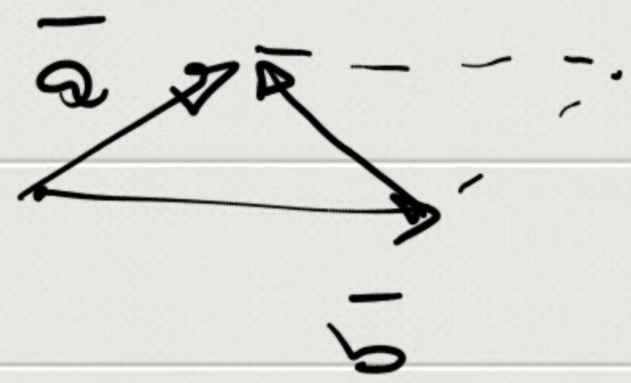
$\bar{c} = \bar{a} - \bar{b} = \bar{a} + (-\bar{b})$



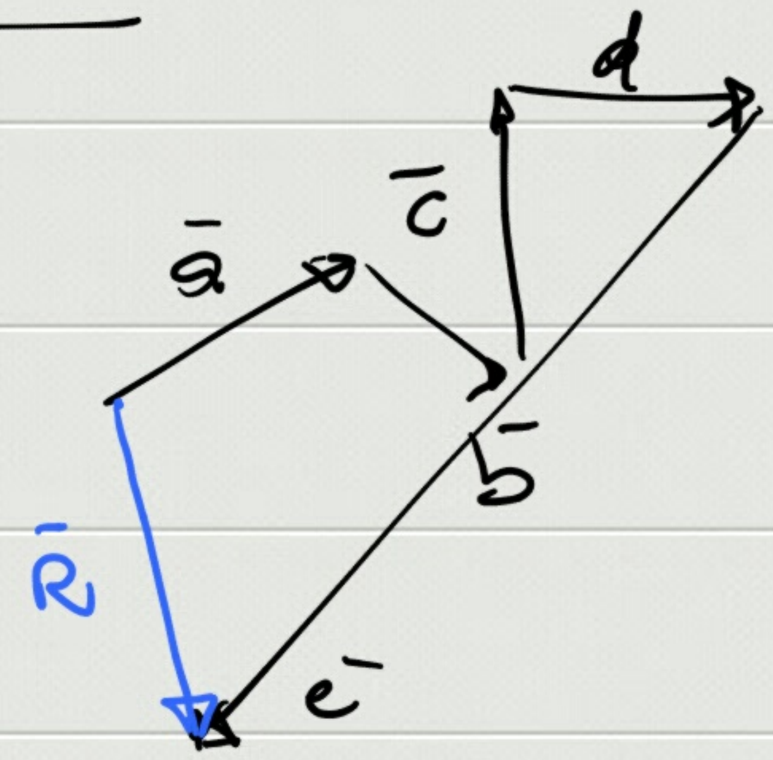








$$\bar{R} = \bar{a} + \bar{b} + \bar{c} + \bar{d}$$



$$(\bar{a} + \bar{b}) + \bar{c} = \bar{a} + (\bar{b} + \bar{c})$$

$$\bar{a}_1 \,//\, \bar{a}_2 \,//\, \bar{a}_3 \,//\, \ldots$$

$$\bar{a}_1 + \bar{a}_2 = |\bar{a}_1|\,\bar{u}_a + |\bar{a}_2|\,\bar{u}_a = (|\bar{a}_1| + |\bar{a}_2|)\,\bar{u}_a$$

$$\sum_i \bar{a}_i = \left(\sum_i |\bar{a}_i|\right) \bar{u}_a$$